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CPE 0411
EMBEDDED SYSTEMS

BoombaStick: Smart Air Drum Sticks

GROUP 3

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I. EXECUTIVE SUMMARY

The BoombaStick: Smart Air Drumming Sticks is a project that presents a new embedded system aimed at providing people with an immersive and portable drumming experience without the use of traditional drum kits. Realizing that standard drums are costly, large, and noisy, the proposed project will offer an approachable, affordable, and easy-to-use option to students, amateur musicians, and hobbyists in the form of smart, movement-sensing drumsticks with digital sound-feedback and learning platforms.

The system is made up of the combination of both hardware and software, which includes an IMU sensor (acceleration and gyroscopes) to track the motion and an Arduino microcontroller to process the data and visual feedback as LED indicators. The mobile application and companion PC reads the motion data and produces live drumming sounds, allowing both performance and learning. Other features are timing feedback to the novices and user customizable sound packs, and MIDI compatibility to the professional music production.

The anticipated benefits are that it will bring about a cost-effective and mobile replacement to the drummers, more involvement in the music education process, and a scalable platform that will serve the needs of both amateur learners and artists. The BoombaStick project is highly feasible based on such measurable parameters as the accuracy of motion tracking, the latency of the device is less than 50 milliseconds, and the level of user satisfaction was more than 80 percent, which confirms that it is an effective educational and entertainment tool.

II. PROBLEM SOLUTION FIT

Who is your customer? AFP personnel, military bands, students, and hobbyists who want a portable, quiet, and affordable way to practice drumming.	Limitations to buy/use your product or service Some users may lack compatible devices or prefer real drum feedback, but portability and low cost outweigh these limits.	How are you going to be different than competition? Unlike other electronic drums, BoombaStick is lightweight, low-cost, and noise-free – ideal for AFT use and local innovation.
Frequent, costly, or urgent problem to solve Traditional drums are loud, bulky, and expensive, limiting access to music practice in camps or small spaces.	Cause of the problem Physical drums require space, produce noise, and cost too much, discouraging everyday practice and morale activities.	Tap into, resemble, or support existing behavior Some users may lack compatible devices or prefer real drum feedback, but portability and low cost outweigh these limits.
Triggers that fit real life, spark associations, make it familiar Drumstick-like design, LED cues, and realistic drum sounds create a familiar feel for both learners and drummers.	“Down-to-earth” solution guess Motion-sensing drumsticks powered by Arduino and IMU sensors that simulate real drumming in the air.	Be where your customers are Demonstrate in AFP camps, schools, ROTC units, and local tech exhibits to reach users directly.
Emotions for stronger message Promotes joy, relaxation, and camaraderie — helping boost morale and creativity among AFP personnel.		

Traditional drum kits are often used in music performances and military ceremonies. However, their high cost, large size, and loud sound make them impractical for many users, especially soldiers in remote camps or those living in small spaces. In the Armed Forces of the Philippines (AFP), keeping troop morale and camaraderie through music is essential to daily life. Unfortunately, traditional instruments are not always easy to transport or maintain in the field.

Additionally, opportunities for music practice or recreation are often limited by space and noise. These challenges highlight the need for a portable, low-noise musical tool that can still provide the experience and rhythm of real drumming.

The BoombaStick: Smart Air Drumming Sticks meets this need by combining embedded systems technology with human creativity. It uses BNO055 IMU sensors to detect movement and an Arduino Nano microcontroller to process signals in real time. With the BoombaStick, users can “play” drums in the air while hearing realistic drum sounds through a connected device. It does not require bulky equipment and can be used anywhere—indoors, outdoors, or even during deployment. This solution offers soldiers and trainees an engaging activity that boosts morale while promoting coordination, focus, and relaxation without disrupting military routines or environments.

Beyond fun, BoombaStick also acts as a practical learning tool for embedded systems and motion-sensing technology. It shows how the same sensors and data-processing concepts can later be used for defense-related innovations, such as gesture-based control systems or wearable monitoring devices. Its compact design, easy calibration, and durable components make it suitable for entertainment and a training platform for AFP personnel interested in affordable local technology solutions.

Thus, the BoombaStick addresses a common limitation—the inability to play full drum kits in restricted environments—by creating an opportunity for both morale boosting and learning. It connects creativity and functionality, demonstrating how a small, student-developed device can help the AFP’s modernization efforts by encouraging innovation, hands-on learning, and a stronger culture of self-reliance in technology.

III. TECHNICAL CONCEPT

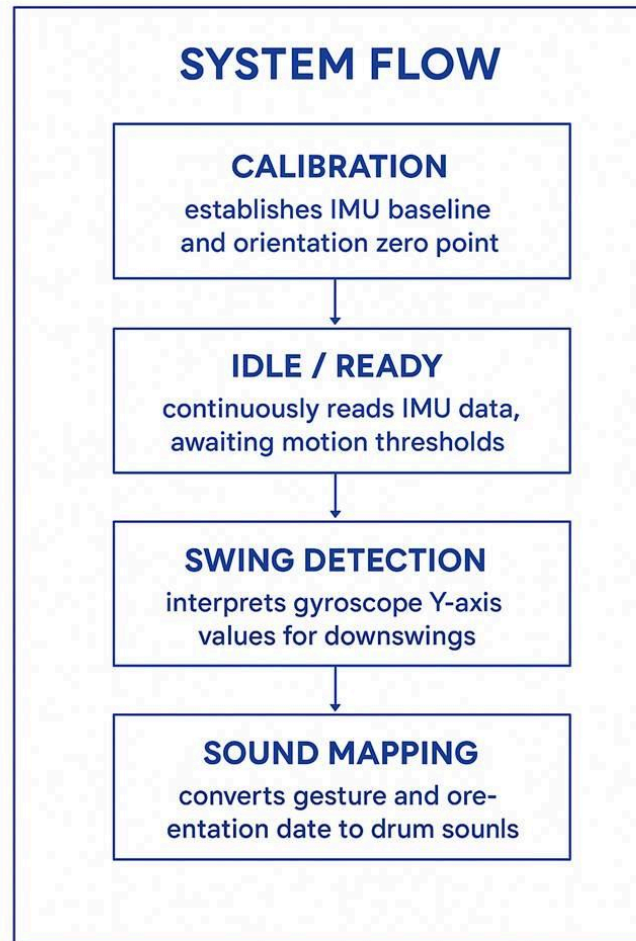
Design Philosophy

Motion-first: prioritize precise, low-latency motion detection to replicate drumming gestures without physical contact. The Invisible Drum Set emphasizes accurate timing and gesture recognition over tactile feedback, using minimal hardware and all-original firmware written in pure C for the Arduino Nano platform. It demonstrates embedded systems mastery through efficient real-time signal processing and sensor fusion using the BNO055 IMU.

Core Functional Modes

- **Calibration:** establishes IMU baseline and orientation zero point for both drumsticks; yaw offset can be reset via button press.
- **Idle / Ready:** continuously reads IMU quaternion and gyroscope data, awaiting motion threshold triggers.
- **Swing Detection:** interprets Y-axis gyroscope values to detect downward stick movements; distinguishes direction and orientation.
- **Sound Mapping:** converts gesture and orientation data to assigned drum sounds (snare, hi-hat, toms, cymbals, kick).
- **Kick Input:** piezo element under shoe or button press triggers kick sound.
- **Recalibration:** resets yaw offsets mid-session to mitigate sensor drift.
- **Error Recovery:** reinitializes sensor communication if I²C or serial errors occur.

System Concept Block Diagram



Sensing & Perception (conceptual)

Sensing uses two BNO055 IMUs—one per drumstick—to provide 9-DOF fused orientation and gyroscope data.

- **Gyroscope (Y-axis):** detects downswings.
- **Pitch and Yaw:** determine which drum/cymbal is being played.

- **Accelerometer:** assists in calibration and normalization.

All sensor data is processed locally on the microcontroller with no raw recording or external computation.

Mobility & Hardware Constraints

- **Controller:** Arduino Nano (ATmega328p, 16 MHz) for compactness and sufficient I/O.
- **Connectivity:** wired via JST connectors for modular assembly and transport safety.
- **Latency goal:** <50 ms end-to-end.
- **Component cost:** ~15,100 ISK total (two IMUs, Nano, piezo, connectors, buttons, resistors).
- **Prototype iterations:** four versions, each improving accuracy, stability, and calibration handling.

Communication & Interface (operational concept)

- **Firmware:**
 - Fully custom C libraries: IMU (BNO055), TWI/I²C, serial, timer, delay, digital input, analog input.
 - Non-blocking millisecond timer interrupts for stable loop timing.
- **Host Software:**
 - Python + Pygame handles serial input and sound playback.
 - Real-time serial codes trigger respective drum samples.
- **User Controls:**
 - Button A: recalibration.
 - Button B or piezo: kick drum trigger.

- Serial feedback for debugging and calibration confirmation.

Safety, Usability, and Reliability

- No physical contact forces or external power risks.
- Safe operating voltage (5V logic).
- Robust wiring and removable connectors prevent tangling or breakage.
- Simple recalibration ensures continuous play despite sensor drift.

Training Integration & Demonstration Scenarios

- **Embedded Systems Education:** illustrates real-time sensor fusion, C-based microcontroller interfacing, and I²C communication.
- **Performance Simulation:** enables silent rhythm practice with realistic timing and motion cues.
- **Future Extensions:** wireless communication, velocity sensitivity, improved piezo durability, and latency optimization through firmware and sound playback pipeline tuning.

IV. MILITARY RELEVANCE

The *BoombaStick: Smart Air Drumming Sticks* aligns with the Armed Forces of the Philippines' (AFP) modernization and innovation framework, particularly under the category of Other Emerging Technologies. While designed primarily as a music and rhythm-learning device, its integration of motion-sensing, inertial measurement, embedded systems, and feedback-based interactivity reflects the type of technological innovation the AFP seeks to foster under the

Self-Reliant Defense Posture (SRDP) and the Revised AFP Modernization Act (Republic Act No. 10349, 2012).

The AFP continues to advance its technological capabilities through the adoption of indigenous and emerging technologies across multiple domains. Recent developments, such as the Philippine Army's deployment of drones and the Philippine Navy's integration of unmanned surface vessels for surveillance and training, demonstrate the military's growing emphasis on innovation and localized research (Philippine News Agency [PNA], 2023; Naval News, 2024). The *BoombaStick* project contributes to this broader modernization effort by showcasing how compact, sensor-based, and interactive technologies can be developed locally for human-centered applications within military environments.

Although the *BoombaStick* does not serve combat or reconnaissance purposes, its relevance to the AFP lies in its potential use for morale, welfare, and recreation (MWR) programs, ceremonial music, and community outreach initiatives. Music has long been integral to military life, improving morale, fostering camaraderie, and reinforcing esprit de corps. In this context, *BoombaStick* provides a modern, low-cost, and portable tool for musical engagement that can be used in AFP camps, bases, or during deployed operations where traditional instruments may not be practical due to space or noise constraints.

Furthermore, the *BoombaStick* highlights the growing defense interest in motion-sensing and embedded systems, technologies also applied in military wearables and human-interface devices. While the system's present form serves creative and recreational purposes, it represents the technological competencies and innovation pathways, in sensor integration, signal

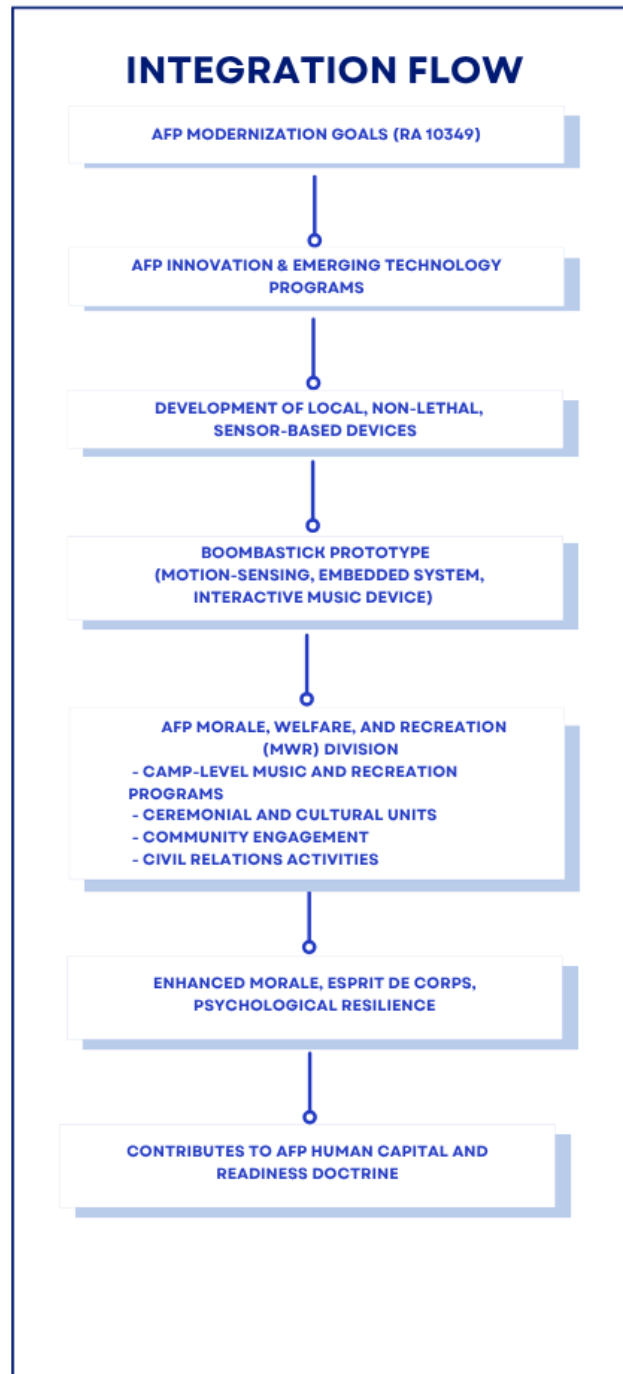
processing, and embedded control, that align with the AFP's future-ready direction toward digitally literate and technically skilled personnel.

Strategic and Doctrinal Fit

Strategically, *BoombaStick* supports the AFP's broader objective of cultivating technological literacy and innovation culture among its personnel. The AFP's modernization roadmap emphasizes *innovation ecosystems* that encourage collaboration with academic institutions and local developers. By integrating a student-developed project utilizing motion sensors and embedded systems, *BoombaStick* embodies the principle of developing indigenous solutions that can later inform more complex defense technologies.

Ideally, *BoombaStick* contributes to the AFP's human capital and morale development doctrine, which values the well-being and cohesion of its personnel as foundations of operational effectiveness. Through music-driven morale initiatives and ceremonial engagement, the device supports psychological resilience, discipline, and cultural identity, non-lethal yet vital aspects of defense readiness.

Integration Flow



To better illustrate the relationship between the *BoombaStick* concept and the AFP's modernization and welfare objectives, the following diagram presents how the device can be integrated into existing military structures and programs. It outlines the process through which *BoombaStick*, as a locally developed motion-sensing and embedded system technology, contributes to the AFP's Other Emerging Technologies initiatives while supporting non-combat domains such as morale, welfare, recreation, and ceremonial engagement. This flow demonstrates the project's alignment with the AFP's commitment to innovation, self-reliance, and human-centered capability development.

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